Environment Diagrams and Lambda Expressions Solutions

Computer Science Mentors 61A

September 16 – September 19, 2025

1 Environment Diagrams

1. Draw the environment diagram that results from running the code. What does the program print?

```
def foo(x):
    def bar():
        print(x)
    return bar

x = foo(3)
y = foo(4)
x()
y()
```

https://tinyurl.com/5n75xdk5

2. Draw the environment diagram that results from running the following code.

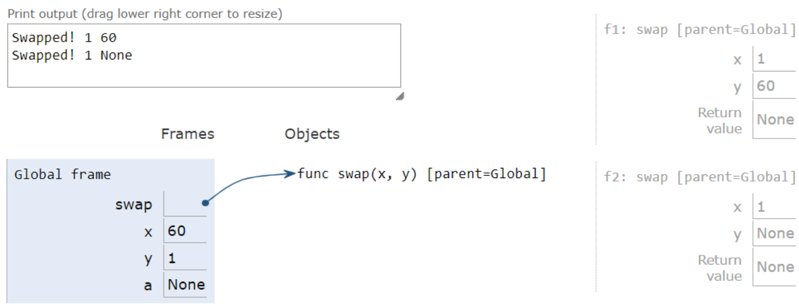
```
a = 1
c = 2
def b(b):
    def d():
         return b + c
    return d()
c = b(a)
a = b(c)
```

Created by Anaisha Das and Tavisha Patel. Worksheet feedback: forms.gle/q67BZrHHfoKbF6V28

3. Give the environment diagram and console output that result from running the following code.

```python
def swap(x, y):
    x, y = y, x
    return print("Swapped!", x, y)

x, y = 60, 1
a = swap(x, y)
swap(a, y)
```



https://tinyurl.com/y68m6qdj

2 Lambda Expressions and Higher Order Functions

1. Express the following expressions using lambdas instead of their named counterparts.

   (a) `square(4)`

   ```python
   (lambda x: x * x)(4)
   ```

   (b) `sum_of_squares(3, 4)`

   ```python
   (lambda x, y: x * x + y * y)(3, 4)
   ```

   (c)
   ```python
   def hello_world():
       return 'Hello, world!'
   hello_world()
   ```

   ```python
   (lambda: 'Hello, world!')()
   ```

2. Make a lambda function, `make_interval()`, that takes in the upper and lower bound of an interval, and returns a function that takes in a value `x` and checks whether `x` is in the interval [`lower`, `upper`], inclusive.

```
>>> make_interval = ______________________________________
>>> in_interval = make_interval(-1, 2)
>>> in_interval(0)
True
>>> in_interval(61)
False
```

```
>>> make_interval = lambda lower, upper: lambda x: x <= upper and x >=
    lower
>>> in_interval = make_interval(-1, 2)
>>> in_interval(0)
True
>>> in_interval(61)
False
```

3. Draw the environment diagram that results from running the code.

```
def a(y):
    d = 1
    b = lambda x: y(x)
    e = lambda x: x(3)
    return e(b)
d = 5
a(lambda x: 4 - x + d)
```

https://goo.gl/9vxEwv